

Introduction to data management

Imagine that you have carefully organized a bookshelf expecting it to stay as you arranged it.

However, the next day, you find that someone has rearranged the books, maybe borrowed the one you were saving, or even replaced it with a different book.

It can be frustrating and confusing when you leave something in one state, but soon find out that it's been changed or removed.

This is especially true for data!

For example, maybe a company collects data about its users.

Some of it's sensitive, and a lot of it only applies to certain departments.

And what if every single employee has access to all of the company's data?

This definitely causes all kinds of problems.

Some employees might mistakenly alter the data or save multiple new versions.

This makes it pretty easy for someone else to begin working with the wrong information.

At the same time, other employees might inadvertently decide to delete some data when they're done with their own projects.

That could be catastrophic!

This business needs a data management plan.

Data management is the process of establishing and communicating a clear plan for collecting, storing, and using data.

A data management plan is created to ensure that procedures for each of these steps are understood by all employees.

In some organizations, the data management plan may also be referred to as **data governance**.

There are **three main reasons** why data management is so important.

First, it ensures that collaboration is **seamless**.

All users can access data using documented procedures, so the data stays within governance and compliance requirements.

Second, data management supports a strong data **security program**.

It helps set parameters to protect from data breaches or data loss.

And **third**, a plan for data management helps with **scalability** by having a clear procedure in place.

As a cloud data analytics professional, you **might create** data management plans.

Or, you could **work within** one that's already been **established**.

Either way, some **key aspects** to consider include **access**, **data types**, **storage**, and **archives**.

Let's dive into each of those.

First, **access**.

A data management plan defines the roles of anyone who will access the data, including each user's level of access.

This can also include the **type of data** they have access to.

It may even be as **granular as certain rows or columns**.

Along the same lines, the plan should also define the data types the organization is allowed to collect, like personally identifiable information, or PII.

For example, maybe only the **marketing department** should have access to user **emails** to send notifications.

Or, only **analysts** have access to user birthdates to segment **reports by age brackets**.

Next, a data management plan should include the types of **storage** allowed.

For example, BigQuery projects and Google Drive folders, and backup plans for outages.

And finally, **archives**.

Procedures for archiving or deleting data should be considered and follow business guidelines and any external regulations.

It's also important to call out any exceptions to archive rules.

For example, litigation data may be exempt from these procedures.

Each of these elements of data management are crucial to protecting information and ensuring user privacy.

While data management plans vary from organization to organization, they should begin by clearly defining the business objectives.

These inform the rest of the plan.

Objectives help define what type of data will be collected and which teams can access it.

They determine how long data should be saved and how it will be used.

And they help prioritize the entire team's goal to effectively share all relevant data management information.

Now, there are some key policies to follow when generating a plan.

First, a retention policy may be created for your organization as a whole or for each individual project.

Retention policies outline how data is archived and deleted.

They may be driven by compliance regulations, legal requirements, and General Data Protection Regulation, or GDPR guidelines.

Next, a data-collection policy is an outline that creates rules around how data is collected and what resources may be used in the process.

The archival policy outlines where and how data is stored once an analysis project is complete.

And lastly, a deletion policy is an outline of when and how data is permanently destroyed.

Of course, if employees don't know about the data management plan and how to access it, that's just as bad as not having one at all.

A business needs to clearly communicate the plan long before there is any data to manage.

And a business should be sure to educate all users about the plan, and the specific permissions and procedures.

That's all for now.

Stay curious.

Hope to continue exploring the cloud with you again soon!